

Competitive Programming Cheat Sheet

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Contents

1 Combinatorics	5		
1.1 Binomial Coefficients	5	1.9.5 Applications	10
1.1.1 Odd numbers in the i -th line	5	2 Geometry	10
1.1.2 Symmetry rule	5	2.1 Distance Metrics	10
1.1.3 Factoring in	5	2.1.1 Euclidean Distance	10
1.1.4 Sum over k (row)	5	2.1.2 Manhattan Distance	11
1.1.5 Sum over n (column)	5	2.1.3 Chebyshev Distance	11
1.1.6 Sum over n and k	5	2.1.4 Relationships Between Metrics	11
1.1.7 Sum of the squares	5	2.1.5 Coordinate Transformation	11
1.1.8 Weighted sum	6	2.2 Point-Line Distance	11
1.1.9 Connection with the Fibonacci numbers	6	2.2.1 Line Equation	11
1.1.10 Pascal's Triangle	6	2.2.2 Distance Formula	11
1.1.11 N -th first terms of P -th column	6	2.2.3 Obtaining Coefficients from Two Points	11
1.1.12 Find if the $\binom{n}{k}$ is odd	6	2.2.4 Signed Distance	11
1.2 Catalan Numbers	6	2.3 Regular Polygons Properties	11
1.2.1 Formulas	6	2.3.1 Angles	11
1.2.2 Applications	6	2.3.2 Relationship	12
1.2.3 Geometric Interpretation	7	2.3.3 Sum of Angles	12
1.2.4 Generalization	7	2.3.4 Special Cases	12
1.2.5 First Catalan Numbers	7	2.4 Trigonometry	12
1.3 Partition	7	2.5 Line	12
1.4 4 fundamental problems of distribution	8	2.5.1 General equation	12
1.4.1 N equal balls in K equal boxes	8	2.5.2 General equation from two points	12
1.4.2 N equal balls in K distinct boxes	8	2.5.3 Line inclination from two points	12
1.4.3 N distinct balls in K equal boxes (Stirling's Number of Second Kind)	8	2.5.4 Check if a point belongs to the line	12
1.4.4 N distinct balls in K distinct boxes	8	2.5.5 Distance from a point to a line	13
1.5 Bell number	8	2.6 Convex and Concave Polygons	13
1.5.1 Bell Triangle	9	2.6.1 Regular polygon circumradius	13
1.5.2 Summation formulas	9	2.6.2 Regular polygon inscribed circle radius	13
1.6 Derangement / Montmort Number / Subfactorial	9	2.6.3 Area of regular polygons	14
1.7 Permutations	9	2.6.4 Sum of internal angle of a regular polygon	14
1.8 Circular permutations	9	2.7 Regular Hexagon	14
1.9 Inclusion-Exclusion Principle	10	2.8 Triangle	15
1.9.1 Two Sets	10	2.8.1 Semiperimeter	15
1.9.2 Three Sets	10	2.8.2 Area	15
1.9.3 General Formula	10	2.8.3 Circumradius	15
1.9.4 Rule	10	2.8.4 Inradius	15
		2.8.5 Length of bisector	15
		2.8.6 Law of sines	15
		2.8.7 Law of cosines	15

2.8.8	Law of tangents	16
2.9	Quadrilaterals	16
2.9.1	Intersection of ... ?	16
2.10	Trapezium	16
2.10.1	Area, knowing base and height	16
2.10.2	Area knowing only sides	16
2.11	Sphere	16
2.11.1	Equation	16
2.11.2	Equation in spherical coordinate system	16
2.11.3	Area	16
2.11.4	Volume	17
2.12	Cube	17
2.12.1	Facial & Body diagonal	17
2.12.2	Area	17
2.12.3	Volume	17
2.12.4	Circumscribed sphere	17
2.12.5	Inscribed Sphere	17
2.12.6	Tangent Sphere	17
2.13	Parallelepiped	17
2.13.1	Area delimited by two vectors	17
2.13.2	Volume	17
2.14	Cylinder	18
2.14.1	Area	18
2.14.2	Volume	18
2.15	Cone	18
2.15.1	Area	18
2.15.2	Volume	18
2.16	Truncated Cone	18
2.16.1	Area	18
2.16.2	Volume	18
2.17	Angles	18
2.17.1	Radians and Degrees Relationship	18
2.17.2	Common Angles	18
2.18	Circle	19
2.18.1	General equation	19
2.18.2	Chord Length	19
2.18.3	Radius from Chord Length	19
2.19	Dot product	19
2.20	Magnitude	19

3	Algebra	19
3.1	Sums	19
3.2	Absolute Value	19
3.2.1	Definition	19
3.2.2	Properties and Theorems	19
3.3	Inequalities	20
3.4	Sophie Germain Identity	20
3.4.1	Verification	20
3.4.2	Alternative Forms	20
3.4.3	Applications	20
4	Graphs	20
4.1	Bipartite Graph	20
4.2	Topological Sorting	21
4.3	Strongly Connected Components	21
4.4	Minimum spanning tree	21
4.4.1	Properties	21
4.5	Eulerian path	21
4.6	Network	21
4.7	Flow network	21
4.7.1	Properties	21
4.8	Prüfer Code	21
4.9	Prüfer's Sequence	22
4.10	Maximum flow	22
4.11	Maximum of disjoint nodes paths	22
4.12	Strongly connected components	22
4.13	Cutpoints / Articulation points	22
4.14	Cayley's Formula	22
4.14.1	Statement	22
4.14.2	Special Cases	22
4.14.3	Prüfer Code	22
4.14.4	Generalizations	23
4.14.5	Applications	23
5	Trees	23
5.1	Centroid	23
5.2	Centroid decomposition	23

6	Number Theory	24			
6.1	Fermat's Theorems and Lemmas	24		6.13	Fundamental theorem of arithmetic 28
6.2	Euler's Theorem	24		6.13.1	LCM and GCD 28
6.2.1	Statement	24		6.14	Taking modulo at the exponent 28
6.2.2	Euler's Totient Function	24		7	Identities
6.2.3	Applications	24		7.1	Geometric sequence 28
6.3	Prime Density	24		7.1.1	Sum of the first N terms 28
6.3.1	Prime Counting Function	24		8	Linear Algebra
6.3.2	Prime Number Theorem	24		8.1	Matrix Multiplication 28
6.3.3	Better Approximations	24		9	Probability Theory
6.3.4	Asymptotic Density	25		9.1	Discrete distributions 29
6.4	Multiplicative Inverse	25		9.1.1	First success distribution 29
6.4.1	Definition	25		9.1.2	Poisson distribution 29
6.4.2	Existence	25		9.2	Continuous distributions 29
6.4.3	Euler totient	25		9.2.1	Uniform distribution 29
6.5	Goldbach's Conjecture	25		9.2.2	Exponential distribution 29
6.6	Linear Diophantine Equations	25		9.2.3	Normal distribution 29
6.6.1	Solution(s)	25		9.3	Markov chains 30
6.7	Wilson's theorem	25		9.3.1	Stationary distribution 30
6.8	Chicken McNugget / Postage Stamp Problem / Froenius Coin Problem	26		9.3.2	Ergodicity 30
6.9	Divisibility rules	26		9.3.3	Absorption 30
6.9.1	7	26		9.4	Monty Hall Problem 30
6.9.2	11	26		9.5	Binomial distribution 30
6.9.3	13	26		9.5.1	Probability Mass Function (PMF) 30
6.9.4	17	26		10	Polynomial
6.9.5	19	26		10.1	Bhaskara 30
6.9.6	23	26		11	Trees
6.10	Chinese Remainder Theorem	27		11.1	Heavy-Light Decomposition 30
6.10.1	Corollary: number modulo a non prime number	27		12	Bitwise
6.10.2	Finding solution	27		12.1	Bitwise Identities 31
6.11	Baby-step giant-step	27		12.1.1	Arithmetic Identities 31
6.11.1	Idea	27		12.1.2	Bit Manipulation Operations 31
6.11.2	Algorithm	27		12.1.3	Useful Tricks 31
6.11.3	Complexity	27		12.2	XOR in range 31
6.12	Carmichael function	27		12.3	Binary to gray code 31
6.12.1	Computation from the prime factorization	28		12.4	Gray code to binary 31
6.12.2	Relation to Euler's totient	28			
6.12.3	Example	28			
6.12.4	First values	28			

13 Game Theory	32
13.1 Impartial Games	32
13.2 Sprague-Grundy Theorem	32
13.3 Nim variation Subtract game	32
13.4 Tweedledum-Tweedledee	32
14 Group Theory	32
14.1 Permutations	32
14.1.1 Odd permutations	32
14.1.2 Even permutations	32
14.1.3 Sum of the number of inversions of every permutation of size N	33
15 Others	33
15.1 Unimodal Functions	33
15.2 Tower of Hanoi	33
15.2.1 Rules	33
15.2.2 Minimum Number of Moves	33
15.2.3 Recursive Strategy	33
15.2.4 Connection to Binary Numbers	34
15.2.5 Generalization: Scrambled Initial State	34

1 Combinatorics

1.1 Binomial Coefficients

Binomial coefficients $\binom{n}{k}$ are the number of ways to select a set of k elements from n different elements without taking into account the order of arrangement of these elements (i.e., the number of unordered sets).

Binomial coefficients are also the coefficients in the expansion of $(a + b)^n$ (so-called binomial theorem):

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{k} a^{n-k} b^k + \dots + \binom{n}{n} b^n$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Recurrence formula** (which is associated with the famous "Pascal's Triangle"):

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

1.1.1 Odd numbers in the i -th line

The number of odd elements in the n -th row in the Pascal's triangle is 2^c , where c is the number of 1 bits in the binary representation of n (popcount)

1.1.2 Symmetry rule

$$\binom{n}{k} = \binom{n}{n-k}$$

1.1.3 Factoring in

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

1.1.4 Sum over k (row)

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

1.1.5 Sum over n (column)

$$\sum_{m=0}^n \binom{m}{k} = \binom{n+1}{k+1}$$

1.1.6 Sum over n and k

$$\sum_{k=0}^m \binom{n+k}{k} = \binom{n+m+1}{m}$$

1.1.7 Sum of the squares

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

1.1.8 Weighted sum

$$1 \binom{n}{1} + 2 \binom{n}{2} + \dots + n \binom{n}{n} = n2^{n-1}$$

1.1.9 Connection with the Fibonacci numbers

$$\binom{n}{0} + \binom{n-1}{1} + \dots + \binom{n-k}{k} + \dots + \binom{0}{n} = F_{n+1}$$

1.1.10 Pascal's Triangle**1.1.11 N-th first terms of P-th column**

$$\binom{p}{p} + \binom{p+1}{p} + \dots + \binom{p+n}{p} = \binom{p+n+1}{p+1} \quad (1)$$

1.1.12 Find if the $\binom{n}{k}$ is odd

In C-style languages, it's just: $!(p \ \& \ (n-p))$.

Or, considering that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

, you can just see if the power of two in each $n!$ is equal to the sum of the powers of two in $r!$ and $(n-r)!$.

1.2 Catalan Numbers

Catalan numbers form a sequence of natural numbers that occur in various counting problems, often involving recursively defined objects.

The n -th Catalan number is denoted as C_n and the sequence begins:

1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, ...

1.2.1 Formulas

Explicit formula:

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$$

Alternative formula:

$$C_n = \binom{2n}{n} - \binom{2n}{n+1}$$

Recurrence relation:

$$C_0 = 1, \quad C_{n+1} = \sum_{i=0}^n C_i C_{n-i}$$

Another recurrence:

$$C_0 = 1, \quad C_n = \frac{2(2n-1)}{n+1} C_{n-1}$$

1.2.2 Applications

The Catalan numbers count:

- **Valid parentheses expressions:** Number of ways to correctly match n pairs of parentheses.
Example: For $n = 3$: $((()))$, $((()))$, $(())()$, $()(())$, $()()()$
- **Binary trees:** Number of different binary trees with n internal nodes.
- **Full binary trees:** Number of full binary trees with $n+1$ leaves (or $2n+1$ nodes total).
- **Rooted trees:** C_{n-1} is the number of different rooted trees with n vertices.
- **Grid paths:** Number of monotonic lattice paths from $(0,0)$ to (n,n) that do not pass above the diagonal $y = x$.
- **Polygon triangulations:** Number of ways to triangulate a convex polygon with $n+2$ sides using non-crossing diagonals.
- **Mountain ranges:** Number of ways to draw n upstrokes and n downstrokes that form a mountain range (never going below the starting height).

1.2.3 Geometric Interpretation

For the grid path problem: we want paths from $(0, 0)$ to (n, n) using only right (R) and up (U) moves, with the constraint that at any point the number of U moves does not exceed the number of R moves.

Total paths without restriction: $\binom{2n}{n}$ (choose n positions for R moves among $2n$ total moves).

Invalid paths: Those that cross the line $y = x + 1$ can be reflected across this line. Any path crossing this line corresponds bijectively to a path from $(0, 0)$ to $(n - 1, n + 1)$, which has $\binom{2n}{n+1}$ paths.

Valid paths (Catalan numbers):

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n}$$

This is known as the **Reflection Principle** or **Cycle Lemma**.

1.2.4 Generalization

For paths from $(0, 0)$ to (m, n) that stay below (or on) the line $y = x + k$:

$$\text{Number of valid paths} = \binom{m+n}{n} - \binom{m+n}{n+k+1}$$

where m is the number of "opening" elements and n is the number of "closing" elements, and k represents how many elements can be "opened" initially.

1.2.5 First Catalan Numbers

n	C_n
0	1
1	1
2	2
3	5
4	14
5	42
6	132
7	429
8	1430
9	4862
10	16796

1.3 Partition

In number theory and combinatorics, a partition of a non-negative integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition. (If order matters, the sum becomes a composition.) For example, 4 can be partitioned in five distinct ways:

$$\begin{aligned} &4 \\ &3 + 1 \\ &2 + 2 \\ &2 + 1 + 1 \\ &1 + 1 + 1 + 1 \end{aligned}$$

The only partition of zero is the empty sum, having no parts.

An individual summand in a partition is called a *part*. The number of partitions of n is given by the partition function $p(n)$. So $p(4) = 5$. The notation $\lambda \vdash n$ means that λ is a partition of n .

1	1	2	3
5	7	11	15
22	30	42	56
77	101	135	176
231	297	385	490
627	792	1002	1255
1575	1958	2436	3010
3718	4565	5604	6842
8349	10143	12310	14883
17977	21637	26015	31185
37338	44583	53174	63261
75175	89134	105558	124754
147273	173525		

Table 1: Values of $a(n)$ for n from 0 to 49

For instance, whenever the decimal representation of n ends in the digit 4 or 9, the number of partitions of n will be divisible by 5.

$P(n, k)$ denotes the number of ways of writing n as a sum of exactly k terms or, equivalently, the number of partitions into parts of which the largest is exactly k . $P(n, k)$ can be computed from the recurrence relation

$$P(n, k) = P(n-1, k-1) + P(n-k, k) \quad (2)$$

With $P(n, k) = 0$ for $k > n$, $P(n, n) = 1$, and $P(n, 0) = 0$. The triangle of $P(k, n)$ is given by :

1.4 4 fundamental problems of distribution

1.4.1 N equal balls in K equal boxes

Considering that no box can be empty, it's given by the partition function when the number of terms is limited by k .

$$P(n, k) = P(n-1, k-1) + P(n-k, k) \quad (3)$$

1.4.2 N equal balls in K distinct boxes

Equivalent to count the number of solutions for the equation:

$$\begin{aligned} x_1 + \dots + x_K &= N \\ \text{where } x_i &> 0, i \in [1, K], N > 0 \end{aligned} \quad (4)$$

It's given by the formula:

$$\binom{N+1}{K-1} \quad (5)$$

If some boxes may be **empty** ($x_i \geq 0$), then it's given by:

$$\binom{N+K-1}{N}$$

1.4.3 N distinct balls in K equal boxes (Stirling's Number of Second Kind)

Also known as **Stirling's Number of Second Kind** let's define $S(N, K)$ as how many distinct ways to distribute N distinguishable balls in K indistinguishable boxes.

Some special cases:

$$\begin{aligned} S(0, 0) &= 1 \\ S(n, 0) &= 0 \\ S(n, k) &= 0 \text{ if } n < k \\ S(n, 1) &= 1 \\ S(n, k) &= 1 \text{ if } n = k \\ S(n, 2) &= 2^{n-1} - 1 \\ S(n, 3) &= \frac{1}{2}(3^{n-1} + 1) - 2^{n-1} \\ S(n, n-1) &= \binom{n}{2} \\ S(n, n-2) &= \binom{n}{3} + 3 \binom{n}{4} \end{aligned} \quad (6)$$

Can be found by the following recurrence relationship.

$$S(n, k) = S(n-1, k-1) + kS(n-1, k) \quad (7)$$

There is also this formula :

$$S(n, k) = \sum_{r=0}^k (-1)^r \frac{(k-r)^n}{r!(k-r)!} \quad (8)$$

1.4.4 N distinct balls in K distinct boxes

$$K^N \quad (9)$$

1.5 Bell number

Count the possible partitions of a set, denoted B_n , where n is an integer greater than or equal to zero.

As an example $B_3 = 5$, because the 3 - element set a, b, c can be partitioned in 5 distinct ways:

$$\begin{aligned} &\{\{a\}, \{b\}, \{c\}\}, \\ &\{\{a\}, \{b, c\}\}, \\ &\{\{b\}, \{a, c\}\}, \\ &\{\{c\}, \{a, b\}\}, \\ &\{\{a, b, c\}\} \end{aligned} \quad (10)$$

The ordering of subsets within the family is not considered.

The first few values, starting from 0.

$$\begin{aligned}
&1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \\
&115975, 678570, 4213597, 27644437, 190899322, \\
&1382958545, 10480142147, 82864869804, 682076806159, \\
&5832742205057, 51724158235372, 474869816156751, \\
&4506715738447323, 44152005855084346, 445958869294805289, \\
&463859033229999353, 49631246523618756274
\end{aligned} \tag{11}$$

1.5.1 Bell Triangle

The Bell numbers can be found using a triangle.

1. Start with the number one. Put this on a row by itself. ($x_{0,1} = 1$)
2. Start a new row with the rightmost element from the previous row as the leftmost number ($x_{i,1} \leftarrow x_{i-1,r}$ where r is the last element of (i-1)-th row)
3. Determine the numbers not on the left column by taking the sum of the number to the left and the number above the number to the left, that is, the number diagonally up and left of the number we are calculating
 $x_{i,j} \leftarrow x_{i,j-1} + x_{i-1,j-1}$
4. Repeat step three until there is a new row with one more number than the previous row (do step 3 until $j = r + 1$)
5. The number on the left hand side of a given row is the Bell number for that row. ($B_i \leftarrow x_{i,1}$)

$$\begin{array}{ccccccc}
1 & & & & & & \\
1 & 2 & & & & & \\
2 & 3 & 5 & & & & \\
5 & 10 & 15 & & & & \\
15 & 20 & 27 & 37 & 52 & &
\end{array} \tag{12}$$

1.5.2 Summation formulas

Using binomial coefficients.

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k \tag{13}$$

Using Stirling numbers of the second kind

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \tag{14}$$

1.6 Derangement / Montmort Number / Subfactorial

In combinatorial mathematics, a derangement is a permutation of the elements of a set in which no element appears in its original position. In other words, a derangement is a permutation that has no fixed points.

The number of derangements of an n -element set is called the n th derangement number or rencontres number, or the subfactorial of n and is sometimes denoted $!n$ or D_n

$$D_n = !n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \tag{15}$$

It also satisfies the following recurrences:

$$!n = n \cdot !(n-1) + (-1)^n \tag{16}$$

$$!n = (n-1) \cdot (!(n-1) + !(n-2)) \tag{17}$$

The first few values, starting from D_0 :

$$\begin{aligned}
&1, 0, 1, 2, 9, 44, 265, 1854, 14833, 133496, \\
&1334961, 14684570, 176214841, 2290792932, 32071101049, \\
&481066515734, 7697064251745, 130850092279664, 2355301661033953, \\
&44750731559645106, 895014631192902121, 18795307255050944540, \\
&413496759611120779881, 9510425471055777937262
\end{aligned} \tag{18}$$

1.7 Permutations

A permutation of a set of r objects is any rearrangement (linear ordering) of the r objects. There are $r!$ (the factorial of r) permutations of a set with r distinct objects.

1.8 Circular permutations

The number of ways to arrange n distinct objects along a fixed (i.e., cannot be picked up out of the plane and turned over) circle is

$$P_n = (n-1)! \tag{19}$$

1.9 Inclusion-Exclusion Principle

The Inclusion-Exclusion Principle calculates the size of the union of multiple sets by alternately adding and subtracting the sizes of their intersections.

1.9.1 Two Sets

For two sets A and B :

$$|A \cup B| = |A| + |B| - |A \cap B| \quad (20)$$

1.9.2 Three Sets

For three sets A , B , and C :

$$|A \cup B \cup C| = |A| + |B| + |C| \quad (21)$$

$$- |A \cap B| - |A \cap C| - |B \cap C| \quad (22)$$

$$+ |A \cap B \cap C| \quad (23)$$

1.9.3 General Formula

For n sets $A_1, A_2, \dots, A_n$:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \sum_{|S|=k} \left| \bigcap_{i \in S} A_i \right| \quad (24)$$

More explicitly:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| \quad (25)$$

$$- \sum_{i < j} |A_i \cap A_j| \quad (26)$$

$$+ \sum_{i < j < k} |A_i \cap A_j \cap A_k| \quad (27)$$

$$- \dots \quad (28)$$

$$+ (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \quad (29)$$

1.9.4 Rule

- **Add** intersections of an **odd** number of sets
- **Subtract** intersections of an **even** number of sets

1.9.5 Applications

Derangements: Count permutations where no element appears in its original position.

The number of derangements D_n of n elements is:

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \approx \frac{n!}{e} \quad (30)$$

Euler's Totient Function: Count integers coprime to n .

For $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$:

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) \quad (31)$$

Counting with constraints: Problems where we need to count objects satisfying "at least one" or "none" of several conditions.

2 Geometry

2.1 Distance Metrics

Different distance metrics are used depending on the problem context. For two points $P = (P_x, P_y)$ and $Q = (Q_x, Q_y)$:

2.1.1 Euclidean Distance

The standard distance in the Euclidean plane:

$$d_2(P, Q) = \sqrt{(P_x - Q_x)^2 + (P_y - Q_y)^2} \quad (32)$$

This is the length of the straight line segment connecting the two points.

2.1.2 Manhattan Distance

Also known as taxicab distance or L_1 distance:

$$d_1(P, Q) = |P_x - Q_x| + |P_y - Q_y| \quad (33)$$

This represents the distance when movement is restricted to horizontal and vertical directions (like moving in a city grid).

2.1.3 Chebyshev Distance

Also known as chessboard distance or L_∞ distance:

$$d_\infty(P, Q) = \max(|P_x - Q_x|, |P_y - Q_y|) \quad (34)$$

This represents the minimum number of moves a king needs to travel between two squares on a chessboard.

2.1.4 Relationships Between Metrics

For any two points P and Q :

$$d_\infty(P, Q) \leq d_2(P, Q) \leq d_1(P, Q) \leq 2 \cdot d_\infty(P, Q) \quad (35)$$

2.1.5 Coordinate Transformation

Manhattan and Chebyshev distances can be converted through a 45° rotation:

Manhattan $\rightarrow$ Chebyshev:

Transform coordinates $(x, y) \mapsto (x + y, x - y)$

Chebyshev $\rightarrow$ Manhattan:

Transform coordinates $(x, y) \mapsto \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$

This transformation is useful when a problem naturally uses one metric but is easier to solve with the other.

2.2 Point-Line Distance

2.2.1 Line Equation

A line in the plane can be represented in the general form:

$$Ax + By + C = 0 \quad (36)$$

where A , B , and C are constants with A and B not both zero.

2.2.2 Distance Formula

The distance from a point $P = (x_0, y_0)$ to the line $Ax + By + C = 0$ is:

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} \quad (37)$$

2.2.3 Obtaining Coefficients from Two Points

Given two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ on the line, the coefficients are:

$$A = y_2 - y_1 \quad (38)$$

$$B = x_1 - x_2 \quad (39)$$

$$C = x_2y_1 - x_1y_2 \quad (40)$$

2.2.4 Signed Distance

The signed distance (positive or negative depending on which side of the line the point is) can be computed as:

$$d_{\text{signed}} = \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \quad (41)$$

Points on opposite sides of the line will have opposite signs. This is useful for determining the orientation or side of a point relative to a line.

2.3 Regular Polygons Properties

A regular polygon has all sides of equal length and all interior angles equal.

2.3.1 Angles

For a regular polygon with n sides:

Exterior angle:

$$\theta_{\text{ext}} = \frac{360^\circ}{n} \quad (42)$$

Interior angle:

$$\theta_{\text{int}} = 180^\circ - \theta_{\text{ext}} = \frac{(n-2) \times 180^\circ}{n} \quad (43)$$

2.3.2 Relationship

The interior and exterior angles at each vertex are supplementary:

$$\theta_{\text{int}} + \theta_{\text{ext}} = 180^\circ \quad (44)$$

2.3.3 Sum of Angles

The sum of all interior angles of an n -sided polygon is:

$$S_{\text{int}} = (n - 2) \times 180^\circ \quad (45)$$

The sum of all exterior angles is always:

$$S_{\text{ext}} = 360^\circ \quad (46)$$

2.3.4 Special Cases

Polygon	n	θ_{ext}	θ_{int}
Equilateral Triangle	3	120°	60°
Square	4	90°	90°
Regular Pentagon	5	72°	108°
Regular Hexagon	6	60°	120°
Regular Octagon	8	45°	135°

2.4 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w \quad (47)$$

$$\cos(v + w) = \cos v \cos w - \sin v \sin w \quad (48)$$

$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w} \quad (49)$$

$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2} \quad (50)$$

$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2} \quad (51)$$

$$(V + W) \tan(v - w)/2 = (V - W) \tan(v + w)/2 \quad (52)$$

where V, W are lengths of sides opposite angles v, w .

$$\begin{aligned} a \cos x + b \sin x &= r \cos(x - \phi) \\ a \sin x + b \cos x &= r \sin(x + \phi) \end{aligned} \quad (53)$$

where $r = \sqrt{a^2 + b^2}$, $\phi = \text{atan2}(b, a)$.

2.5 Line

2.5.1 General equation

$$ax + by + c = 0 \quad (54)$$

Note that a same line can have multiple representations (just multiply the equation by any real number), so to make each line have a single equation divide everything by a , or b if a is zero.

2.5.2 General equation from two points

Let P and Q be the points that define the line.

$$\begin{aligned} a &= P_y - Q_y \\ b &= Q_x - P_x \\ c &= P \times Q = P_x Q_y - P_y Q_x \end{aligned}$$

2.5.3 Line inclination from two points

Let P and Q be two points that belongs to the line, such that $P_x < Q_x$ the inclination m or angular coefficient is given by:

$$m = \frac{Q_y - P_y}{Q_x - P_x} \quad (55)$$

2.5.4 Check if a point belongs to the line

Let r be a line such that $ax + by + c = 0$ and P a point. $P \in r$ if and only if :

$$aP_x + bP_y + c = 0$$

2.5.5 Distance from a point to a line

The distance from a point P and a line r is defined as the shortest distance possible between every point that belongs to r and P . Such distance will be the distance from P and the intersection between r and the orthogonal projection from P to r , and can be found by:

$$\frac{|aP_x + bP_y + c|}{\sqrt{a^2 + b^2}}$$

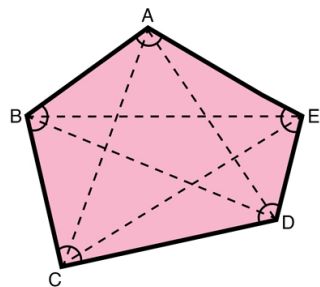
The coordinates of the point Q are given by:

$$Q_x = \frac{b(bP_x - aP_y) - ac}{a^2 + b^2}$$

$$Q_y = \frac{a(-bP_x + aP_y) - bc}{a^2 + b^2}$$

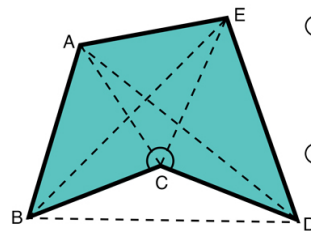
2.6 Convex and Concave Polygons

Convex Polygon



- ① All interior angles measure $< 180^\circ$
- ② All diagonals lie inside the closed figure

Concave Polygon



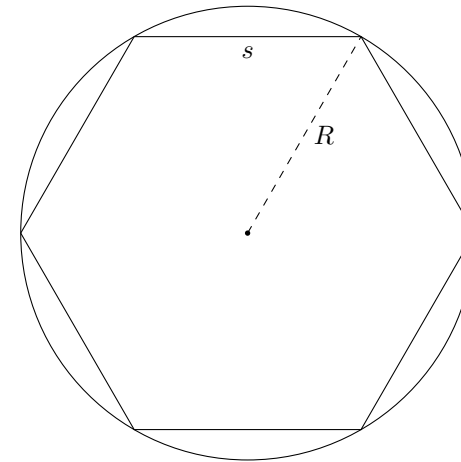
- ① At least one interior angle measure $> 180^\circ$ and $< 360^\circ$
- ② At least one diagonal lie outside the closed figure

Figure 2: Convex Polygon

Figure 3: Concave Polygon

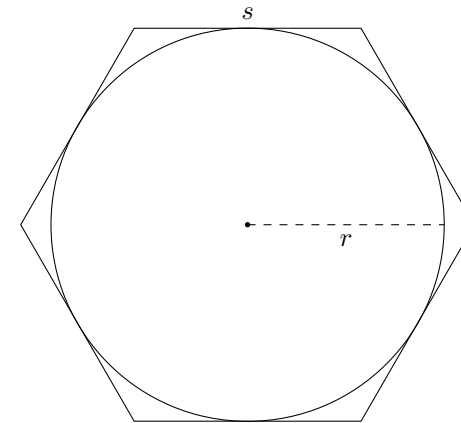
Figure 4: Two Types of Polygons

2.6.1 Regular polygon circumradius



$$R = \frac{s}{2} \csc \frac{\pi}{n}$$

2.6.2 Regular polygon inscribed circle radius



$$r = R \cos \frac{\pi}{n}$$

2.6.3 Area of regular polygons

- Let n be the number of sides of the regular polygon, the area can be found using one of the values below:

- the length of one of the sides (s)
- apothem, the radius of the inscribed circle (r)
- the radius of the circumscribed circle (R)

$$A = \frac{1}{2}nrs = \frac{1}{4}ns^2 \cot \frac{\pi}{n} = nr^2 \tan \frac{\pi}{n} = \frac{1}{2}nR^2 \sin \frac{2\pi}{n}$$

2.6.4 Sum of internal angle of a regular polygon

A regular polygon with n sides have $(n - 2)180$ degrees as sum of it's internal angle.

2.7 Regular Hexagon

A regular hexagon is a polygon with six equal sides and six equal angles (each 120°). Let s be the side length of the regular hexagon.

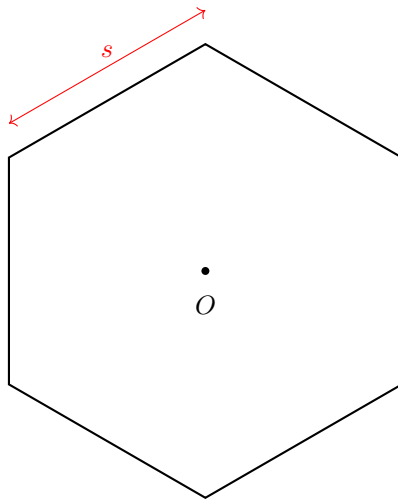


Figure 5: Regular Hexagon with side length s

Area:

$$A = \frac{3\sqrt{3}}{2}s^2 \approx 2.598s^2$$

Diagonals:

A regular hexagon has two types of diagonals:

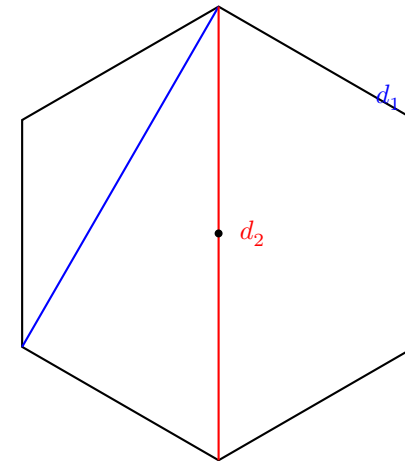


Figure 6: Hexagon diagonals: short diagonal d_1 (blue) and long diagonal d_2 (red)

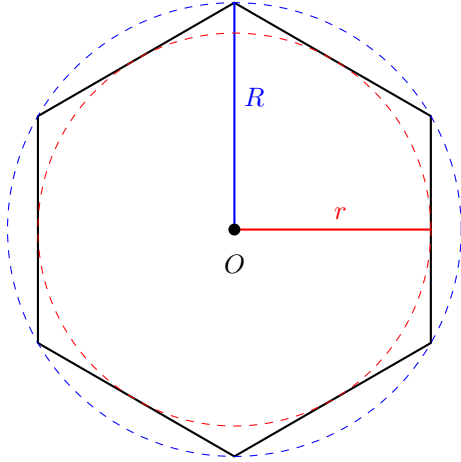
Short diagonal (connecting vertices separated by one vertex):

$$d_1 = \sqrt{3}s$$

Long diagonal (connecting opposite vertices):

$$d_2 = 2s$$

Circumradius and Inradius:

Figure 7: Circumradius R (blue) and inradius r (red)

Circumradius (radius of circumscribed circle):

$$R = s$$

Inradius (radius of inscribed circle, or apothem):

$$r = \frac{\sqrt{3}}{2}s \approx 0.866s$$

Relationship between circumradius and inradius:

$$r = \frac{\sqrt{3}}{2}R$$

Area from radius:

Area in terms of circumradius R :

$$A = \frac{3\sqrt{3}}{2}R^2$$

Area in terms of inradius r :

$$A = 2\sqrt{3}r^2$$

Internal angles:

The sum of all internal angles is $(6 - 2) \times 180^\circ = 720^\circ$

Each internal angle of a regular hexagon is 120°

2.8 Triangle

Let the length of the sides of the triangle be a, b, c .

2.8.1 Semiperimeter

Let p be the semiperimeter defined as:

$$p = \frac{a + b + c}{2} \quad (56)$$

2.8.2 Area

Let A be the area defined as:

$$\sqrt{p(p-a)(p-b)(p-c)} \quad (57)$$

2.8.3 Circumradius

$$R = \frac{abc}{4A} \quad (58)$$

2.8.4 Inradius

$$r = \frac{A}{p} \quad (59)$$

2.8.5 Length of bisector

$$s_a = \sqrt{bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right]} \quad (60)$$

2.8.6 Law of sines

$$\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R} \quad (61)$$

2.8.7 Law of cosines

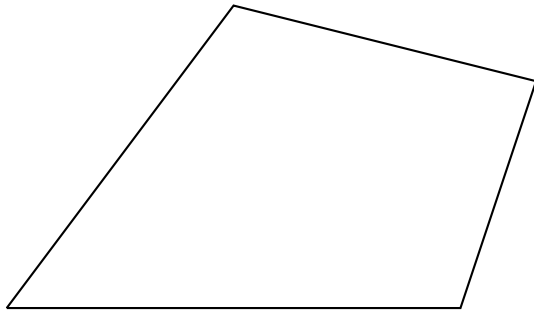
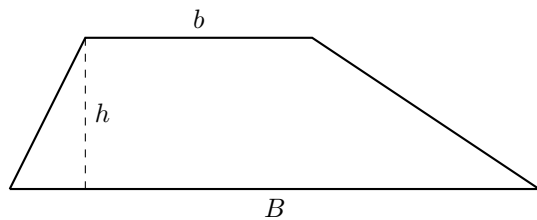
$$a^2 = b^2 + c^2 - 2bc \cos \alpha \quad (62)$$

2.8.8 Law of tangents

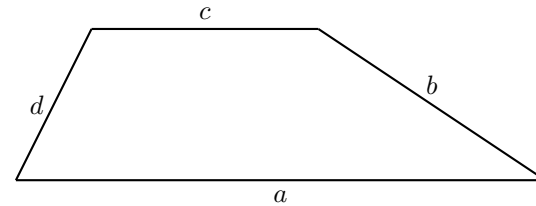
$$\frac{a+b}{a-b} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}} \quad (63)$$

2.9 Quadrilaterals

Let its sides length be a, b, c, d

2.9.1 Intersection of ... ?**2.10 Trapezium****2.10.1 Area, knowing base and height**

$$A = \frac{(B+b)h}{2}$$

2.10.2 Area knowing only sides

$$e = \frac{d^2 - b^2 + a^2 - 2ac + c^2}{2a - 2c}$$

$$h = \sqrt{d^2 - e^2}$$

$$A = \frac{h(a+c)}{2}$$

2.11 Sphere**2.11.1 Equation**

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2$$

2.11.2 Equation in spherical coordinate system

r is the radius, θ is an angle that goes from 0 to 2π , and φ is an angle that goes from 0 to π .

$$x = x_0 + r \cos \theta \sin \varphi$$

$$y = y_0 + r \sin \theta \sin \varphi$$

$$z = z_0 + r \cos \varphi$$

2.11.3 Area

$A = 4\pi r^2$, where r is the radius, comes from :

$$A = \int_0^\pi \int_0^{2\pi} r^2 \sin(\varphi) d\theta d\varphi$$

2.11.4 Volume

$$V = \frac{4}{3}\pi r^3$$

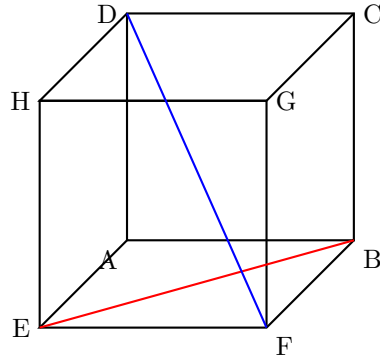
comes from :

$$V = \int_0^R \int_0^\pi \int_0^{2\pi} r^2 \sin(\varphi) d\theta d\varphi dR$$

2.12 Cube**2.12.1 Facial & Body diagonal**

Facial diagonal join two vertices at the same face.

Body diagonal join two vertices from opposite faces.



$$\text{Facial diagonal} = L\sqrt{2}$$

$$\text{Body diagonal} = L\sqrt{3}$$

2.12.2 Area

$$A = 6L^2$$

2.12.3 Volume

$$V = L^3$$

2.12.4 Circumscribed sphere

Pass through the 8 vertices, radius equal to $L(\frac{\sqrt{3}}{2})$.

2.12.5 Inscribed Sphere

Tangent to the 6 faces, radius equal to $\frac{L}{2}$.

2.12.6 Tangent Sphere

Tangent to the edges, radius equal to $\frac{L}{\sqrt{2}}$.

2.13 Parallelepiped

Let it be defined by three linear independent vectors

$$\vec{u}, \vec{v}, \vec{w}$$

2.13.1 Area delimited by two vectors

$$A = |\vec{u} \times \vec{v}|$$

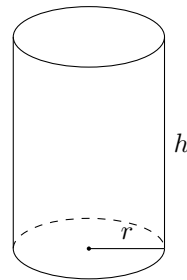
2.13.2 Volume

$$V = |(\vec{u} \times \vec{v}) \cdot \vec{w}|,$$

$$(\vec{u} \times \vec{v}) \cdot \vec{w} = \det \begin{bmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{bmatrix}$$

or if you know the sides a, b, c , and the angles α, β, γ :

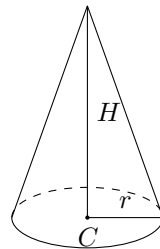
$$V = abc\sqrt{1 + 2\cos\alpha\cos\beta\cos\gamma - \cos^2\alpha - \cos^2\beta - \cos^2\gamma}$$

2.14 Cilinder**2.14.1 Area**

$$A = 2\pi r h + 2\pi r^2 = 2\pi r(h + r),$$

2.14.2 Volume

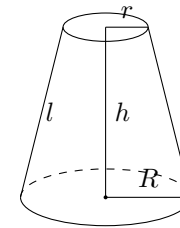
$$A = \pi r^2 h$$

2.15 Cone**2.15.1 Area**

$$A = \pi r^2 + \pi r \sqrt{r^2 + H^2}$$

2.15.2 Volume

$$V = \frac{1}{3} \pi r^2 H$$

2.16 Truncated Cone**2.16.1 Area**

$$l = \sqrt{h^2 + R^2}$$

$$A = \pi(R + r)l + \pi(R^2 + r^2)$$

2.16.2 Volume

$$V = \frac{1}{3} \pi h (R^2 + Rr + r^2),$$

2.17 Angles**2.17.1 Radians and Degrees Relationship**

A radian is the angle subtended by an arc equal in length to the radius of the circle. The relationship between radians and degrees is:

$$\text{radians} = \text{degrees} \times \frac{\pi}{180}$$

$$\text{degrees} = \text{radians} \times \frac{180}{\pi}$$

One full rotation is 2π radians or 360° .

2.17.2 Common Angles

What is wrong with this table ?

θ	$30^\circ \left(\frac{\pi}{6}\right)$	$45^\circ \left(\frac{\pi}{4}\right)$	$60^\circ \left(\frac{\pi}{3}\right)$
$\sin \theta$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos \theta$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan \theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

2.18 Circle

2.18.1 General equation

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

2.18.2 Chord Length

A chord is a line segment with both endpoints on the circle. To find the length of a chord:

If the central angle is θ (in radians) and the radius is r , the chord length is:

$$c = 2r \sin\left(\frac{\theta}{2}\right)$$

Alternatively, if the perpendicular distance from the center to the chord is d , and the radius is r :

$$c = 2\sqrt{r^2 - d^2}$$

2.18.3 Radius from Chord Length

Given a chord of length c and the perpendicular distance d from the center to the chord, the radius can be found using:

$$r = \sqrt{d^2 + \left(\frac{c}{2}\right)^2}$$

Alternatively, if the central angle θ (in radians) and chord length c are known:

$$r = \frac{c}{2 \sin\left(\frac{\theta}{2}\right)}$$

2.19 Dot product

The dot product of vectors $\mathbf{u}$ and $\mathbf{v}$ in n dimensions is given by:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 \cdot v_1 + u_2 \cdot v_2 + \dots + u_n \cdot v_n$$

2.20 Magnitude

The magnitude of a vector $\mathbf{v}$ in n dimensions is given by:

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

The dot product of two Euclidean vectors $\mathbf{u}$ and $\mathbf{v}$ is defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = |\mathbf{u}| \cdot |\mathbf{v}| \cdot \cos(\theta)$$

3 Algebra

3.1 Sums

$$\begin{aligned} c^a + c^{a+1} + \dots + c^b &= \frac{c^{b+1} - c^a}{c - 1}, c \neq 1 \\ 1 + 2 + 3 + \dots + n &= \frac{n(n+1)}{2} \\ 1^2 + 2^2 + 3^2 + \dots + n^2 &= \frac{n(2n+1)(n+1)}{6} \\ 1^3 + 2^3 + 3^3 + \dots + n^3 &= \frac{n^2(n+1)^2}{4} \\ 1^4 + 2^4 + 3^4 + \dots + n^4 &= \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \end{aligned} \quad (64)$$

3.2 Absolute Value

3.2.1 Definition

The absolute value of x , denoted by $|x|$ is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \quad (65)$$

3.2.2 Properties and Theorems

$$|x| < a \iff -a < x < a, \text{ where } a > 0 \quad (66)$$

$$|x| \leq a \iff -a \leq x \leq a, \text{ where } a > 0 \quad (67)$$

$$|x| > a \iff (x > a) \vee (x < -a), \text{ where } a > 0 \quad (68)$$

$$|x| \geq a \iff (x \geq a) \vee (x \leq -a), \text{ where } a > 0 \quad (69)$$

$$|ab| = |a| \cdot |b|, \text{ where } a, b \in \mathbb{R} \quad (70)$$

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}, \text{ where } a, b \in \mathbb{R}, \text{ and } b \neq 0 \quad (71)$$

$$|a + b| \leq |a| + |b|, \text{ where } a, b \in \mathbb{R} \quad (72)$$

$$|a - b| \leq |a| + |b|, \text{ where } a, b \in \mathbb{R} \quad (73)$$

$$|a| - |b| \leq |a - b|, \text{ where } a, b \in \mathbb{R} \quad (74)$$

3.3 Inequalities

$$\begin{aligned} x &< a \\ x &< b \\ x &< \min(a, b) \end{aligned} \quad (75)$$

$$\begin{aligned} x &\leq a \\ x &\leq b \\ x &\leq \min(a, b) \end{aligned} \quad (76)$$

$$\begin{aligned} x &> a \\ x &> b \\ x &> \max(a, b) \end{aligned} \quad (77)$$

$$\begin{aligned} x &\geq a \\ x &\geq b \\ x &\geq \max(a, b) \end{aligned} \quad (78)$$

3.4 Sophie Germain Identity

The Sophie Germain Identity is a factorization formula for the sum of two fourth powers:

$$a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab) \quad (79)$$

3.4.1 Verification

The identity can be verified by expanding the right-hand side:

$$\begin{aligned} &(a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab) \\ &= (a^2 + 2b^2)^2 - (2ab)^2 \\ &= a^4 + 4a^2b^2 + 4b^4 - 4a^2b^2 \\ &= a^4 + 4b^4 \end{aligned}$$

3.4.2 Alternative Forms

The identity can also be written as:

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2) \quad (80)$$

Or in terms of complex numbers:

$$a^4 + 4b^4 = (a^2 + 2b^2 + 2abi)(a^2 + 2b^2 - 2abi) \quad (81)$$

3.4.3 Applications

The Sophie Germain Identity is useful for:

- Factoring expressions of the form $x^4 + 4y^4$
- Proving that certain numbers are composite
- Solving Diophantine equations
- Number theory problems involving fourth powers

4 Graphs

4.1 Bipartite Graph

Bipartite Graph is a special graph with the following characteristics: the set of vertices V can be partitioned into two disjoint sets V_1 and V_2 and all undirected edges $(u, v) \in E$ have the property that $u \in V_1$ and $v \in V_2$. This makes a Bipartite Graph free from odd-length cycle. Not that by this definition is possible to have isolated vertices.

4.2 Topological Sorting

Definition: You are given a **directed** graph with n vertices and m edges. You have to find an order of the vertices, so that every edge leads from the vertex with a smaller index to a vertex with a larger one.

Topological order can be non-unique !

A Topological order **may not exist** at all. It only exists, if the directed graph **contains no cycles**. Otherwise because there is a contradiction: if there is a cycle containing the vertices a and b , then a needs to have a smaller index than b (since you can reach b from a) and also a bigger one (as you can reach a from b). Every acyclic directed graph contains at least one topological order.

4.3 Strongly Connected Components

You are given a directed graph G with vertices V and edges E . It is possible that there are loops and multiple edges. Let's denote n as number of vertices and m as number of edges in G .

Strongly connected component is a maximal subset of vertices C such that any two vertices of this subset are reachable from each other, i.e. for any $u, v \in C$:

$$u \mapsto v, v \mapsto u$$

where $\mapsto$ means reachability, i.e. existence of the path from first vertex to the second.

The most important property of the condensation graph is that it is a **DAG**. Indeed, suppose that there is an edge between C and C' , let's prove that there is no edge from C' to C . Suppose that $C' \mapsto C$. Then there are two vertices $u' \in C$ and $v' \in C'$ such that $v' \mapsto u'$. But since u and u' are in the same strongly connected component then there is a path between them; the same for v and v' . As a result, if we join these paths we have that $v \mapsto u$ and at the same time $u \mapsto v$. Therefore u and v should be at the same strongly connected component, so this is contradiction. This completes the proof.

4.4 Minimum spanning tree

4.4.1 Properties

- A minimum spanning tree of a graph is unique, if the weight of all the edges are distinct. Otherwise, there may be multiple minimum spanning trees. (Specific algorithms typically output one of the possible minimum spanning trees).

- Minimum spanning tree is also the tree with minimum product of weights of edges. (It can be easily proved by replacing the weights of all edges with their logarithms)

4.5 Eulerian path

A Eulerian path is a path in a graph that passes through all of its edges exactly once. A Eulerian cycle is a Eulerian path that is a cycle.

An Eulerian cycle exists if and only if the degrees of all vertices are even. And an Eulerian path exists if and only if the number of vertices with odd degrees is two (or zero, in the case of the existence of a Eulerian cycle).

4.6 Network

A **network** is a directed graph G with vertices V and edges E combined with a function c , which assigns each edge $e \in E$ a non-negative integer value, the *capacity* of e .

4.7 Flow network

Is a **network** with two vertices labeled as **source** and **sink**.

4.7.1 Properties

- The flow of an edge cannot exceed the capacity

$$f(e) \leq c(e)$$

- And the sum of the incoming flow of a vertex u has to be equal to the sum of the outgoing flow of u except in the source and sink vertices.
- The source vertex s only has an outgoing flow, and the sink vertex t has only incoming flow.

4.8 Prüfer Code

The Prüfer code is a way of encoding a labeled tree with n vertices using a sequence of $n - 2$ integers in the interval $[0; n - 1]$. This encoding also acts as a bijection between all spanning trees of a complete graph and the numerical sequences.

The Prüfer code is constructed as follows. We will repeat the following procedure $n - 2$ times: we select the leaf of the tree with the smallest

number, remove it from the tree, and write down the number of the vertex that was connected to it. After $n - 2$ iterations there will only remain 2 vertices, and the algorithm ends.

Thus the Prüfer code for a given tree is a sequence of $n - 2$ numbers, where each number is the number of the connected vertex, i.e. this number is in the interval $[0, n - 1]$.

The algorithm for computing the Prüfer code can be implemented easily with $O(n \log n)$ time complexity, simply by using a data structure to extract the minimum (for instance set or priority_queue in C++), which contains a list of all the current leafs.

After constructing the Prüfer code two vertices will remain. One of them is the highest vertex $n - 1$, but nothing else can be said about the other one.

Each vertex appears in the Prüfer code exactly a fixed number of times - its degree minus one. This can be easily checked, since the degree will get smaller every time we record its label in the code, and we remove it once the degree is 1. For the two remaining vertices this fact is also true.

4.9 Prüfer's Sequence

The Prüfer sequence is a bijection between labeled trees with n vertices and sequences with $n - 2$ numbers from 1 to n .

To get the sequence from the tree:

- While there are more than 2 vertices, remove the leaf with smallest label and append it's neighbour to the end of the sequence.

To get the tree from the sequence:

- The degree of each vertex is 1 more than the number of occurrences of that vertex in the sequence. Compute the degree d , then do the following: for every value x in the sequence (in order), find the vertex with smallest label y such that $d(y) = 1$ and add an edge between x and y , and also decrease their degrees by 1. At the end of this procedure, there will be two vertices left with degree 1; add an edge between them.

4.10 Maximum flow

4.11 Maximum of disjoint nodes paths

Given a directed graph count the maximum number of distinct paths, such that there isn't two paths that share a node.

Run a flow on the graph but uses two nodes to represent a single node in the original graph, let I_u be the "input node of u ", and O_u be the "output node of u ". for each edge (u, v) in the original graph put an edge (O_u, I_v) in the flow graph. Also add an edge between I_u, O_u for every u that way even that multiple edges "chooses I_u ", as between I_u and O_u there is only 1 of capacity, O_u will have at most one edge leaving it.

4.12 Strongly connected components

4.13 Cutpoints / Articulation points

Definition 1 (Cutpoints / Articulation points) In an *undirected graph* an *articulation point* (or *cut vertex*) is defined as a vertex which, when removed along with associated edges, makes the graph disconnected (or more precisely, increases the number of connected components in the graph).

4.14 Cayley's Formula

Cayley's Formula counts the number of distinct labeled trees on n vertices.

4.14.1 Statement

The number of labeled trees with n vertices is:

$$n^{n-2} \quad (82)$$

4.14.2 Special Cases

n	n^{n-2}	Trees
1	$1^{-1} = 1$	1
2	$2^0 = 1$	1
3	$3^1 = 3$	3
4	$4^2 = 16$	16
5	$5^3 = 125$	125
6	$6^4 = 1296$	1296

4.14.3 Prüfer Code

Cayley's Formula can be proven using the Prüfer code, which establishes a bijection between:

- Labeled trees with n vertices

- Sequences of length $n - 2$ with elements from $\{1, 2, \dots, n\}$

There are exactly n^{n-2} such sequences.

4.14.4 Generalizations

Forests: The number of labeled forests with n vertices and k trees is:

$$k \cdot n^{n-k-1} \quad (83)$$

Spanning trees: For a complete graph K_n , the number of spanning trees equals n^{n-2} (which is Cayley's Formula).

4.14.5 Applications

Cayley's Formula is used in:

- Counting problems involving labeled trees
- Graph enumeration
- Network design and reliability analysis
- Combinatorial optimization

5 Trees

5.1 Centroid

A centroid of a tree is defined as a node such that when the tree is rooted at it, no other nodes have a subtree of size greater than $\frac{N}{2}$.

- if $N \geq 3$ the centroid is never a leaf
- Every tree have a centroid.
- There is at most two centroids (mostly only one)
- All centroids are adjacent
- Let $s(v)$ be the sum of the distance from v to every other node, essentially $\sum_{u \neq v} \text{dist}(u, v)$, then if, c is the centroid, it implies that $s(c)$ is the smallest one among every other node.

5.2 Centroid decomposition

The centroid decomposition of a tree is another tree defined recursively as:

- Its root is the centroid of the original tree.
- Its children are the centroid of each tree resulting from the removal of the centroid from the original tree.

Properties:

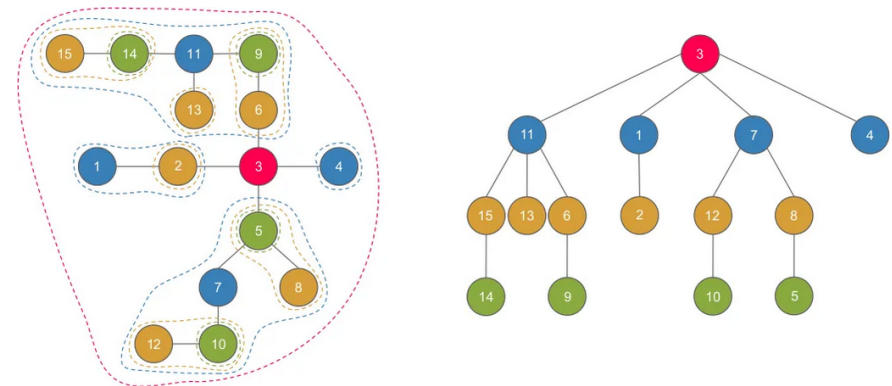
- The tree formed contains all the N nodes.
- The height of the centroid decomposition is $O(\log N)$.
- Let A and B be two arbitrary nodes, let C be the $LCA(A, B)$ in the centroid decomposition tree.

C lies on the path from A to B in the original tree. So every path in the tree can be decomposed in A to C , and C to B .

As the height of the tree is $O(\log N)$ the LCA can be found in $O(\log N)$ (just like with binary lifting)

- We can represent all possible paths in the original tree (there are n^2 paths) as a concatenation of two paths in a $O(n \log n)$ set composed of paths like this: $(u, \text{fa}(u)), (\text{fa}(u), \text{fa}(\text{fa}(u))), \dots$ for all possible u .

Example:



6 Number Theory

6.1 Fermat's Theorems and Lemmas

Let p be a prime number and $a, b \in \mathbb{Z}$:

$$a^p \equiv a \pmod{p} \quad (84)$$

$$a^{p-1} \equiv 1 \pmod{p} \quad (85)$$

$$(a+b)^p \equiv a^p + b^p \pmod{p} \quad (86)$$

$$a^{-1} \equiv a^{p-2} \pmod{p} \quad (87)$$

$$a^{-1} \equiv a^{p-2} \pmod{p} \quad (\gcd(a, p) = 1) \quad (88)$$

6.2 Euler's Theorem

6.2.1 Statement

Let a and n be positive integers with $\gcd(a, n) = 1$. Then:

$$a^{\varphi(n)} \equiv 1 \pmod{n} \quad (89)$$

where $\varphi(n)$ is Euler's totient function, which counts the number of integers from 1 to n that are coprime to n .

6.2.2 Euler's Totient Function

The totient function $\varphi(n)$ has several important properties:

For prime powers:

$$\varphi(p^k) = p^{k-1}(p-1) = p^k - p^{k-1} \quad (90)$$

Multiplicative property: If $\gcd(m, n) = 1$, then:

$$\varphi(mn) = \varphi(m) \cdot \varphi(n) \quad (91)$$

General formula: For $n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$:

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) \quad (92)$$

6.2.3 Applications

Modular exponentiation:

When computing $a^b \pmod{n}$ with $\gcd(a, n) = 1$, we can reduce the exponent:

$$a^b \equiv a^{b \bmod \varphi(n)} \pmod{n} \quad (93)$$

This is particularly useful when b is very large.

Modular multiplicative inverse:

If $\gcd(a, n) = 1$, then:

$$a^{-1} \equiv a^{\varphi(n)-1} \pmod{n} \quad (94)$$

Computing $a^{b^c} \pmod{n}$:

When $\gcd(a, n) = 1$:

$$a^{b^c} \equiv a^{b^c \bmod \varphi(n)} \pmod{n} \quad (95)$$

First compute $e = b^c \bmod \varphi(n)$, then compute $a^e \bmod n$.

6.3 Prime Density

6.3.1 Prime Counting Function

The prime counting function $\pi(n)$ denotes the number of prime numbers less than or equal to n .

6.3.2 Prime Number Theorem

The Prime Number Theorem provides an asymptotic approximation for $\pi(n)$:

$$\pi(n) \sim \frac{n}{\ln(n)} \quad (96)$$

More precisely:

$$\lim_{n \rightarrow \infty} \frac{\pi(n)}{n/\ln(n)} = 1 \quad (97)$$

6.3.3 Better Approximations

A more accurate approximation is given by the logarithmic integral:

$$\pi(n) \sim \text{Li}(n) = \int_2^n \frac{dt}{\ln(t)} \quad (98)$$

For practical purposes, the following approximation is also useful:

$$\pi(n) \approx \frac{n}{\ln(n) - 1} \quad \text{for } n \geq 17 \quad (99)$$

6.3.4 Asymptotic Density

The probability that a random integer near n is prime is approximately $\frac{1}{\ln(n)}$. The n -th prime number p_n can be approximated by:

$$p_n \sim n \ln(n) \quad (100)$$

6.4 Multiplicative Inverse

6.4.1 Definition

A multiplicative inverse of an integer a modulo m is an integer x such that:

$$a \cdot x \equiv 1 \pmod{m}$$

6.4.2 Existence

A multiplicative inverse exists if and only if a and m are coprime, i.e., $\gcd(a, m) = 1$.

6.4.3 Euler totient

If a and m are coprime, the multiplicative inverse can be computed using Euler's theorem:

$$x \equiv a^{\varphi(m)-1} \pmod{m}$$

If you want to find the smallest i such that $x^i \equiv 1 \pmod{m}$, we can test every divisor of $\varphi(m)$ until we find the smallest i , such that $x^i \equiv 1 \pmod{m}$.

6.5 Goldbach's Conjecture

"Every pair number greater than 2 can be written as the sum of two primes"

Valid for every integer in range from 4 to 10^{18} , but without proof

For an odd x number it can be written as the sum of two primes if $x - 2$ is also prime, or three primes, 3 and the two primes that results in $x - 3$.

6.6 Linear Diophantine Equations

A Linear Diophantine Equation (in two variables) is an equation of the general form:

$$ax + by = c$$

Where a, b, c are given integers, and x, y are unknown integers. If $a = b = 0$, we have infinite solutions if $c = 0$, and 0 otherwise.

6.6.1 Solution(s)

Let $g = \gcd(a, b)$ such that $ax_g + by_g = g$, then we only have a solution if and only if $g \mid c$, and if it have a solution it have infinite.

The solutions will be of the form :

$$\begin{aligned} x_0 &= x_g \cdot \frac{c}{g}, y_0 = y_g \cdot \frac{c}{g}. \\ a \cdot x_0 + b \cdot y_0 &= c \end{aligned} \quad (101)$$

With the initial solution, we can can find every solution, with :

$$x = x_0 + k \cdot \frac{b}{g}, y = y_0 - k \cdot \frac{a}{g} \quad (102)$$

To find the solution that minimize $x + y$ we use the fact that:

$$\begin{aligned} x' &= x + k \cdot \frac{b}{g}, \\ y' &= y - k \cdot \frac{a}{g}. \end{aligned}$$

Note that

$x + y$ change as follows:

$$x' + y' = x + y + k \cdot \left(\frac{b}{g} - \frac{a}{g} \right) = x + y + k \cdot \frac{b - a}{g}$$

If $a < b$, we need to select smallest possible value of k . If $a > b$, we need to select the largest possible value of k . If $a = b$, all solution will have the same sum $x + y$.

6.7 Wilson's theorem

Wilson's theorem states that a natural number $n > 1$ is a **prime number** if and only if the product of all the positive integers less than n is one less than a multiple of n .

That is :

$$(n - 1)! \equiv -1 \pmod{n} \quad (103)$$

In other words, any integer $n > 1$ is a prime number if, and only if, $(n - 1)! + 1$ is divisible by n

6.8 Chicken McNugget / Postage Stamp Problem / Froenius Coin Problem

For any two **relative prime positive integers**, m, n , the greatest integer that cannot be written in the form $am + bn$ for **nonnegative** a and b is $mn - m - n$.

6.9 Divisibility rules

6.9.1 7

To check whether a number is divisible by 7, follow this rule:

1. Take the number in question.
2. Remove the last digit (units) of the number.
3. Double the value removed in the previous step.
4. Subtract the doubled value from the remaining number.
5. If the result of the subtraction is divisible by 7, the original number is divisible by 7.

Example:

Suppose we want to check whether the number 413 is divisible by 7.

1. Remove the last digit (3) and double it, obtaining 6.
2. Subtract 6 from the remaining number ($41 - 6 = 35$).

6.9.2 11

$$n \text{ is divisible by } 11 \iff \sum_{i=1}^k a_{2i-1} - \sum_{i=1}^j a_{2i} \text{ is divisible by } 11$$

where a_i is the i -th digit of the number n , k is the number of odd-position digits, and j is the number of even-position digits.

Example:

Suppose we want to check whether the number $n = 7923$ is divisible by 11.

$$k = 2, \quad j = 2$$

$$\text{Sum of odd-position digits: } 7 + 3 = 10$$

$$\text{Sum of even-position digits: } 9 + 2 = 11$$

$$\text{Subtraction: } 10 - 11 = -1$$

Since -1 is not divisible by 11, the number 7923 is not divisible by 11.

6.9.3 13

$$13|x \equiv 13|4 \cdot (x \% 10) + \lfloor x/10 \rfloor \quad (104)$$

In other words, 13 divides x if four times the last digit added to the number without that digit is divisible by 13.

6.9.4 17

$$17|x \equiv 17|\lfloor x/10 \rfloor - 5 \cdot (x \% 10) \quad (105)$$

In other words, 17 divides x if the difference between the number without its last digit and five times that last digit is divisible by 17.

6.9.5 19

$$19|x \equiv 19|\lfloor x/10 \rfloor + 2 \cdot (x \% 10) \quad (106)$$

In other words, 19 divides x if twice the last digit of x added to the remaining number of x is divisible by 19.

6.9.6 23

$$23|x \equiv 23|x/10 + 7 \cdot (x \% 10) \quad (107)$$

6.10 Chinese Remainder Theorem

Let $m = m_1 \cdot m_2 \cdots m_k$, where m_i are pairwise coprime. In addition to m_i , we are also given a system of congruences

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{cases}$$

where a_i are some given constants. The original form of CRT then states that the given system of congruences always has one and exactly one solution modulo m .

6.10.1 Corollary: number modulo a non prime number

A consequence of the CRT is that the equation

$$x \equiv a \pmod{m}$$

is equivalent to the system of equations

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{cases}$$

(As above, assume that $m = m_1 m_2 \cdots m_k$ and m_i are pairwise coprime).

6.10.2 Finding solution

Let $M = m_1 m_2 \cdots m_k$, and $o_i = \frac{M}{m_i}$. If the numbers a_i satisfy $a_i o_i \equiv 1 \pmod{m_i}$, for every $1 \leq i \leq k$, then a solution for x is $\sum_{i=1}^k a_i o_i a_i \pmod{M}$

6.11 Baby-step giant-step

A meet-in-the-middle algorithm for solving the discrete logarithm problem: given a cyclic group G of order n , a generator α and an element β , find an integer x such that

$$\alpha^x = \beta. \quad (108)$$

6.11.1 Idea

Let $m = \lceil \sqrt{n} \rceil$. Any exponent $0 \leq x < n$ can be written as

$$x = im + j, \quad 0 \leq i < m, \quad 0 \leq j < m. \quad (109)$$

Substituting into $\alpha^x = \beta$ gives

$$\alpha^{im+j} = \beta \iff \alpha^j = \beta(\alpha^{-m})^i. \quad (110)$$

We precompute every *baby step* α^j and store it in a hash table, then take *giant steps* of size m multiplying β by α^{-m} until a match is found.

6.11.2 Algorithm

1. $m \leftarrow \lceil \sqrt{n} \rceil$.
2. For all j with $0 \leq j < m$, compute α^j and store the pair (α^j, j) in a table.
3. Compute α^{-m} .
4. $\gamma \leftarrow \beta$.
5. For all i with $0 \leq i < m$:
 - (a) If γ is in the table as α^j , return $x = im + j$.
 - (b) Otherwise, $\gamma \leftarrow \gamma \cdot \alpha^{-m}$.

6.11.3 Complexity

$$\text{Time } O(\sqrt{n}), \quad \text{Space } O(\sqrt{n}), \quad (111)$$

using a hash table for the lookups. It is a generic algorithm that works for any finite cyclic group and does not require knowing the exact order in advance. For groups of composite order, Pohlig-Hellman is usually faster.

6.12 Carmichael function

The Carmichael function $\lambda(n)$ is the smallest positive integer m such that

$$a^m \equiv 1 \pmod{n} \quad (112)$$

holds for every integer a coprime to n . In other words, it is the exponent of the multiplicative group of integers modulo n .

6.12.1 Computation from the prime factorization

For a prime power p^r ,

$$\lambda(p^r) = \begin{cases} \frac{1}{2}\varphi(p^r) & \text{if } p = 2 \text{ and } r \geq 3, \\ \varphi(p^r) & \text{otherwise,} \end{cases} \quad (113)$$

where $\varphi(p^r) = p^{r-1}(p-1)$. For a general $n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$,

$$\lambda(n) = \text{lcm}(\lambda(p_1^{r_1}), \lambda(p_2^{r_2}), \dots, \lambda(p_k^{r_k})). \quad (114)$$

6.12.2 Relation to Euler's totient

Since $\lambda(n)$ is the exponent of the multiplicative group modulo n while $\varphi(n)$ is its order,

$$\lambda(n) \mid \varphi(n). \quad (115)$$

6.12.3 Example

For $n = 8$ we have $\varphi(8) = 4$ but $\lambda(8) = 2$ (the special case for powers of 2). Indeed $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$.

6.12.4 First values

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\lambda(n)$	1	1	2	2	4	2	6	2	6	4	10	2	12	6	4	4

6.13 Fundamental theorem of arithmetic

Every integer greater than 1 can be represented uniquely as a product of prime numbers, up to the order of the factors.

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} \quad (116)$$

6.13.1 LCM and GCD

$$\begin{aligned} a &= p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} \\ b &= p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k} \\ (a, b) &= p_1^{\min \alpha_1, \beta_1} p_2^{\min \alpha_2, \beta_2} \dots p_k^{\min \alpha_k, \beta_k} \\ [a, b] &= p_1^{\max \alpha_1, \beta_1} p_2^{\max \alpha_2, \beta_2} \dots p_k^{\max \alpha_k, \beta_k} \end{aligned} \quad (117)$$

6.14 Taking modulo at the exponent

If $\gcd(a, m) = 1$ then:

$$a^m \equiv a^{n \bmod \varphi(m)} \pmod{m} \quad (118)$$

7 Identities

$$\begin{aligned} \sum_{i=1}^n i &= \frac{n(n+1)}{2} \\ \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6} \\ \sum_{i=1}^n i^3 &= \frac{n^2(n+1)^2}{4} = \left(\sum_{i=1}^n i \right)^2 \end{aligned}$$

7.1 Geometric sequence

$$a, ar, ar^2, ar^3, \dots, ar^{n-1} \quad (119)$$

r should not be 0

7.1.1 Sum of the first N terms

$$\sum_{k=0}^{n-1} ar^k = a \frac{1-r^n}{1-r} \quad (120)$$

8 Linear Algebra

8.1 Matrix Multiplication

$$\begin{matrix} \begin{matrix} n \\ \text{---} \\ \text{m} \end{matrix} & \cdot & \begin{matrix} p \\ \text{---} \\ n \end{matrix} & = & \begin{matrix} p \\ \text{---} \\ m \end{matrix} \\ A & \cdot & B & = & C \end{matrix}$$

Let $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times c}$. The product $C = AB \in \mathbb{R}^{n \times c}$ is defined as:

$$c_{ij} = \sum_{k=1}^m a_{ik} b_{kj} \quad (121)$$

For $i = 1, \dots, n$ and $j = 1, \dots, m$.

For matrix multiplication, the number of **columns in the first matrix** must be equal to the number of **rows in the second matrix**. The result matrix has the number of **rows of the first** and the number of **columns of the second matrix**.

Matrix multiplication is associative and distributive, but not commutative: $AB \neq BA$ in general.

9 Probability Theory

Let X be a **discrete random** variable with probability $p_X(x)$ of assuming the value x .

It will then have an **expected value** (mean) $\mu = \mathbb{E}(X) = \sum_x x p_X(x)$

Variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation.

If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y ,

$$V(aX + bY) = a^2 V(X) + b^2 V(Y).$$

9.1 Discrete distributions

9.1.1 First success distribution

The number of trials needed to get the first success in independent yes/no experiments, each which yields success with probability p is $\text{Fs}(p)$, $0 \leq p \leq 1$.

$$p(k) = p(1-p)^{k-1}, \quad k = 1, 2, \dots$$

$$\mu = \frac{1}{p}, \quad \sigma^2 = \frac{1-p}{p^2}$$

9.1.2 Poisson distribution

The number of events occurring in a fixed period of time t if these events occur with a known average rate κ and independently of the time since the last event is $\text{Po}(\lambda)$, $\lambda = t\kappa$.

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$\mu = \lambda, \quad \sigma^2 = \lambda$$

9.2 Continuous distributions

9.2.1 Uniform distribution

If the probability density function is constant between a and b and 0 elsewhere it is $\text{U}(a, b)$, $a < b$.

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$

9.2.2 Exponential distribution

The time between events in a Poisson process is $\text{Exp}(\lambda)$, $\lambda > 0$.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda}, \quad \sigma^2 = \frac{1}{\lambda^2}$$

9.2.3 Normal distribution

Most real random values with mean μ and variance σ^2 are well described by $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

9.3 Markov chains

A *Markov chain* is a discrete random process with the property that the next state depends only on the current state. Let $X_1, X_2, \dots$ be a sequence of random variables generated by the Markov process. Then there is a transition matrix $\mathbf{P} = (p_{ij})$, with $p_{ij} = \Pr(X_n = i | X_{n-1} = j)$, and $\mathbf{p}^{(n)} = \mathbf{P}^n \mathbf{p}^{(0)}$ is the probability distribution for X_n (i.e., $p_i^{(n)} = \Pr(X_n = i)$), where $\mathbf{p}^{(0)}$ is the initial distribution.

9.3.1 Stationary distribution

is a stationary distribution if $\pi = \mathbf{P} \pi$. If the Markov chain is *irreducible* (it is possible to get to any state from any state), then $\pi_i = \frac{1}{\mathbb{E}(T_i)}$ where $\mathbb{E}(T_i)$ is the expected time between two visits in state i . π_j / π_i is the expected number of visits in state j between two visits in state i . For a connected, undirected and non-bipartite graph, where the transition probability is uniform among all neighbors, π_i is proportional to node i 's degree.

9.3.2 Ergodicity

A Markov chain is *ergodic* if the asymptotic distribution is independent of the initial distribution. A finite Markov chain is ergodic iff it is irreducible and *aperiodic* (i.e., the gcd of cycle lengths is 1). $\lim_{k \rightarrow \infty} \mathbf{P}^k = \mathbf{1}\pi$.

9.3.3 Absorption

A Markov chain is an **A-chain** if the states can be partitioned into two sets **A** and **G**, such that all states in **A** are absorbing ($p_{ii} = 1$), and all states in **G** leads to an absorbing state in **A**. The probability for absorption in state $i \in \mathbf{A}$, when the initial state is j , is $a_{ij} = p_{ij} + \sum_{k \in \mathbf{G}} a_{ik} p_{kj}$. The expected time until absorption, when the initial state is i , is $t_i = 1 + \sum_{k \in \mathbf{G}} p_{ki} t_k$.

9.4 Monty Hall Problem

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

Savant's response was that the contestant should switch to the other door. By the standard assumptions, the switching strategy has a $\frac{2}{3}$ probability of winning the car, while the strategy of keeping the initial choice has only a $\frac{1}{3}$ probability.

9.5 Binomial distribution

9.5.1 Probability Mass Function (PMF)

The probability of obtaining exactly k successes in n independent Bernoulli trials is defined by the formula:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

The distribution satisfies a linear recurrence relation based on the total number of trials. To achieve k successes in n trials, one must have either had $k-1$ successes in $n-1$ trials followed by a success, or k successes in $n-1$ trials followed by a failure:

$$P(n, k) = p \cdot P(n-1, k-1) + (1-p) \cdot P(n-1, k)$$

with boundary conditions $P(0, 0) = 1$ and $P(n, k) = 0$ for $k < 0$ or $k > n$.

10 Polynomial

10.1 Bhaskara

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (122)$$

11 Trees

11.1 Heavy-Light Decomposition

Heavy-Light Decomposition (HLD) is a technique to decompose a tree into a set of disjoint paths. This technique is particularly useful to deal with problems which require us to do some path-queries in a tree which seemingly complicated but easy enough to be solved for a line-graph. The idea is to decompose the tree into several paths (line-graph) of disjoint vertices. Then, each path-query in the original tree might be able to be answered by queries in one or more of those paths.

An edge (a, b) is heavy if and only if $\text{size}(b) \geq \text{size}(a)/2$; otherwise, it is light

12 Bitwise

12.1 Bitwise Identities

12.1.1 Arithmetic Identities

Sum using AND and OR:

$$a + b = (a \& b) + (a | b) \quad (123)$$

Sum using XOR and AND:

$$a + b = (a \oplus b) + 2(a \& b) \quad (124)$$

XOR using AND and OR:

$$a \oplus b = \sim (a \& b) \& (a | b) \quad (125)$$

12.1.2 Bit Manipulation Operations

Let x denote the bit position (0-indexed from the right).

Set (turn on) the x -th bit:

$$n | (1 \ll x) \quad (126)$$

Flip (toggle) the x -th bit:

$$n \oplus (1 \ll x) \quad (127)$$

Clear (turn off) the x -th bit:

$$n \& \sim (1 \ll x) \quad (128)$$

Check if the x -th bit is set:

$$n \& (1 \ll x) \neq 0 \quad (129)$$

12.1.3 Useful Tricks

Clear all trailing ones:

$$n \& (n + 1) \quad (130)$$

Example: 0011 0111 $\rightarrow$ 0011 0000

Set the rightmost zero bit:

$$n | (n + 1) \quad (131)$$

Example: 0011 0101 $\rightarrow$ 0011 0111

Isolate the least significant bit (LSB):

$$n \& (-n) \quad (132)$$

Example: 0011 0100 $\rightarrow$ 0000 0100

Remove the least significant bit:

$$n \& (n - 1) \quad (133)$$

Example: 0011 0100 $\rightarrow$ 0011 0000

Check if n is a power of 2:

$$(n \& (n - 1)) = 0 \quad \text{and} \quad n \neq 0 \quad (134)$$

Powers of 2 have exactly one bit set in their binary representation.

12.2 XOR in range

Let us introduce the notation $XOR(l, r) = l \oplus (l + 1) \oplus \dots \oplus r$.

Then

$$XOR(0, x) = \begin{cases} x & \text{if } x \equiv 0 \pmod{4} \\ 1 & \text{if } x \equiv 1 \pmod{4} \\ x + 1 & \text{if } x \equiv 2 \pmod{4} \\ 0 & \text{if } x \equiv 3 \pmod{4} \end{cases} \quad (135)$$

Then $XOR(l, r)$ can be found as $XOR(0, r) \oplus XOR(0, l - 1)$.

12.3 Binary to gray code

$$\begin{aligned} G_n &= B_n \\ G_{n-1} &= B_n \oplus B_{n-1} \\ G_i &= B_i \oplus B_{i-1} \\ &\dots \\ G_1 &= B_2 \oplus B_1 \end{aligned}$$

12.4 Gray code to binary

$$\begin{aligned} B_n &= G_n \\ B_{n-1} &= B_n \oplus G_{n-1} = G_n \oplus G_{n-1} \\ &\dots \\ B_1 &= B_2 \oplus G_1 = G_n \oplus G_1 \end{aligned}$$

13 Game Theory

13.1 Impartial Games

To be considered a impartial game following rules must be true:

1. The available moves win/lose depends only on the state of the game, in other words, the only difference between the two players is that one of them moves first
2. Additionally, we assume that the game has perfect information, i.e. no information is hidden from the players (they know the rules and the possible moves).
3. It is assumed that the game is finite, i.e. after a certain number of moves, one of the players will end up in a losing position — from which they can't move to another position. On the other side, the player who set up this position for the opponent wins. Understandably, there are no draws in this game.

Such games can be completely described by a directed acyclic graph: the vertices are game states and the edges are transitions (moves). A vertex without outgoing edges is a losing vertex (a player who must make a move from this vertex loses).

Since there are no draws, we can classify all game states as either winning or losing. Winning states are those from which there is a move that causes inevitable defeat of the other player, even with their best response. Losing states are those from which all moves lead to winning states for the other player. Summarizing, a state is winning if there is at least one transition to a losing state and is losing if there isn't at least one transition to a losing state. Our task is to classify the states of a given game.

13.2 Sprague-Grundy Theorem

The Sprague-Grundy Theorem states that every impartial game is equivalent to a pile of a certain size in Nim. In other words, every impartial game can be solved as Nim by finding their corresponding game.

Basically, for a game situation A and its SG function value $g(A)$:

1. $g(A) = 0$ if and only if A is a must-lose situation. Otherwise, $g(A) \in \mathbb{Z}^*$
2. If A can be divided into n sub-situations $x_1, x_2, \dots, x_n$, then $g(A) = g(x_1) \oplus g(x_2) \oplus \dots \oplus g(x_n)$

3. If A can be converted to situation B_1 or B_2 or ... or B_n by only one operation, then $g(A) = \text{mex}(g(B_1), g(B_2), \dots, g(B_n))$.

13.3 Nim variation Subtract game

Work just like nim but instead remove any number of objects you can remove at most K , this game can be seen as a nim game but before computing the num-sum you need to take the size of each pile module $K + 1$, the optimal way to play it is by taking K at each turn.

13.4 Tweedledum-Tweedledee

The Tweedledum-Tweedledee strategy is a copycat strategy in symmetrical games where the second player mirrors the first player's moves to guarantee a win.

14 Group Theory

14.1 Permutations

A permutation is an arrangement of elements. A permutation of N elements can be represented by an arrangement of the numbers $1, 2, \dots, N$ in some order. Eg. 5, 1, 4, 2, 3.

14.1.1 Odd permutations

A permutation is called odd if it can be expressed as a product of odd number of transpositions.

A sorted permutation is an even permutation

14.1.2 Even permutations

A permutation is called even if it can be expressed as a product of even number of transpositions.

14.1.3 Sum of the number of inversions of every permutation of size N

$$\frac{N(N-1)N!}{4} \quad (136)$$

Some of the first values starting at 1:

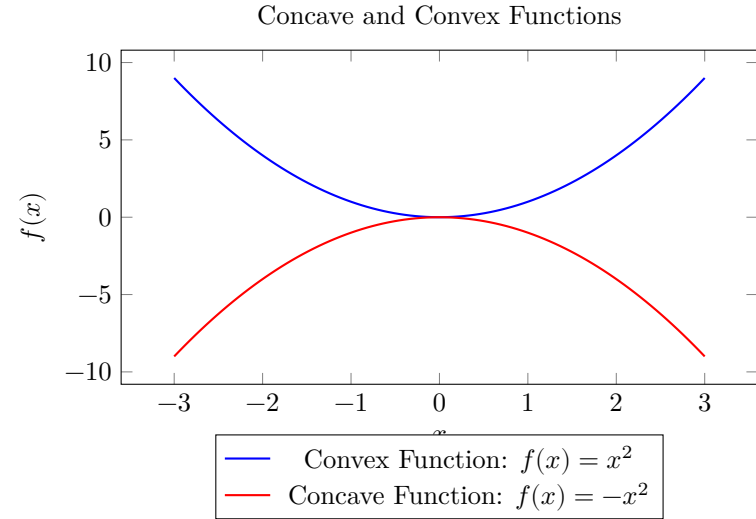
0, 1, 9, 72, 600, 5400, 52920, 564480, 6531840, 81648000, 1097712000,
 15807052800, 242853811200, 3966612249600, 68652904320000,
 1255367393280000, 24186745110528000, 489781588488192000,
 10400656084955136000, 231125690776780800000, 5364548928029491200000,
 129822084058313687040000, 3270280117468949544960000,
 85621879439187042631680000, 2326681506499647897600000000,
 65534862433073415782400000000, 19109965885484208042147840000000,
 57623897131613919634784256000000, 1794877684729159496772354048000000,
 57692497009151555253397094400000000,
 19118099870963670551211933696000000000,
 652564475595559954814700670156800000000,
 22923958513663380348168033219379200000000,
 828128001306089615077570200050073600000000,
 30741115199998781165758287729131520000000000,
 1171778979388188835024198261675130880000000000,
 45833297793783729004232212006664404992000000000,
 18384245003951029078364253927117611335680000000000,
 75574153110836527643762243846340234706944000000000

15 Others

15.1 Unimodal Functions

By unimodal function, we mean one of two behaviors of the function:

- The function strictly increases first, reaches a maximum (at a single point or over an interval), and then strictly decreases.
- The function strictly decreases first, reaches a minimum, and then strictly increases.



15.2 Tower of Hanoi

The Tower of Hanoi is a classic mathematical puzzle consisting of three pegs and n disks of different sizes.

15.2.1 Rules

- Only one disk can be moved at a time
- A disk can only be placed on top of a larger disk or on an empty peg
- All disks start on one peg and must be moved to another peg

15.2.2 Minimum Number of Moves

The minimum number of moves required to transfer n disks from one peg to another is:

$$T(n) = 2^n - 1 \quad (137)$$

15.2.3 Recursive Strategy

To move n disks from peg A to peg C using auxiliary peg B :

1. Move $n - 1$ disks from A to B (using C as auxiliary)

2. Move the largest disk from A to C
3. Move $n - 1$ disks from B to C (using A as auxiliary)

15.2.4 Connection to Binary Numbers

The sequence of moves corresponds to the binary representation of integers from 1 to $2^n - 1$:

- Disk i (counting from 1 at the top) moves when the i -th bit changes in

the binary counter

- The solution follows Gray code sequence

15.2.5 Generalization: Scrambled Initial State

If the disks start in an arbitrary valid configuration (not necessarily all on one peg), the problem becomes more complex but can still be solved recursively by considering the position of each disk.